Computing Infrastructures
July 4, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona): ....

Last Name: ....
(LAST NAME IN CAPITAL LETTERS)

First Name: ....
(FIRST NAME IN CAPITAL LETTERS)

Exam Duration: 1hour and 30min

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True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

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Question 1 All hardware accelerators used in the datacenter context use seamless programming interfaces and software frameworks, and can be easily interchanged.

A False B True

Question 2 Warehouse-scale computers are single servers that are designed to be used in large-scale data centers.

A False B True

Question 3 Docker and Kubernetes are two technology examples supporting para-virtualization.

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Question 4 East-west traffic refers to network traffic that travels between servers within the same datacenter.

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Question 5 The average seek time of an HDD can be estimated as one-third of the time it takes the actuator arm to move from the innermost to the outermost track.

A True B False

Question 6 Replicating data across availability zones helps ensure data durability in case of local failures.

A True B False

Question 7 3-tier datacenter networks consist of three distinct layers named core, aggregation, and access.

A True B False

Question 8 Data centers using HDDs typically ignore redundancy features, assuming disk failures are very rare.

A True B False

Question 9 Cloud-native applications are designed to run in the cloud and often use microservices and containers.

A True B False

Question 10 Open-loop cooling exploits favourable outdoor conditions to reduce mechanical cooling load.

A True B False

Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11
Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 8 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?

Question 12
A datacenter hosts a critical service that relies on a system composed of three components:

- Component A is a central control unit (e.g., a network controller or orchestrator) that is mandatory for system operation. If A fails, the service becomes unavailable.
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Question 13

Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 420 days and a Mean Time To Repair (MTTR) of 4 days. What is the maximum number of disks that can be included while ensuring that the overall storage ($MTTF_{RAID1+0}$) is at least 8 years

Question 14

An Internet based system includes initially a Web server, an application server, and a DBMS. After monitoring the system for $T = 1800s$ you get the following data:

- Number of completions $C = 600$
- Web server utilization $U_{Web} = 75\%$
- Application server utilization $U_{App} = 50\%$
- DBMS utilization $U_{DBMS} = 60\%$

What is the service demand for the three servers (i.e. D_{Web}, D_{App}, D_{DBMS})?

Question 15

Given the system described at Question 14, you expect the incoming load to grow to $\lambda = 5 req/s$. Determine the minimum number of servers at each physical layer in a way that the utilization of each server is at most 55% (assume that the servers of the same type evenly distribute their load).

Question 16

Assume you upgraded the system according to your answer to Question 15. What is the maximum throughput that the new system can achieve?

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

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Answer Sheets (Page 1)

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Answer Sheets (Page 2)

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Answer Sheets (Page 3)

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True/False Questions

Question 01 : ☒A ☐B

Question 02 : ☒A ☐B

Question 03 : ☒A ☐B

Question 04 : ☐A ☒B

Question 05 : ☒A ☐B

Question 06 : ☒A ☐B

Question 07 : ☒A ☐B

Question 08 : ☐A ☒B

Question 09 : ☒A ☐B

Question 10 : ☒A ☐B

Exercises

Question 11 : ....32 SERVERS............

Question 12 : ...0,9957 $= A_V A$ ...... $A_{V\,SYS} = 0{,}9584$

Question 13 : ....15.1 ⇒ 14 DISKS............

Question 14 : $D_{WEB} = 2{,}25$ SEC $D_{APP} = 1{,}5$ SEC $D_{DBMS} = 1{,}8$ SEC

Question 15 : $N_{WEB} = 21$ ...... $N_{APP} = 14$ ...... $N_{DBMS} = 17$

Question 16 : ....$X_{MAX} = 9{,}\bar{3}$ REQ/SEC............

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A True B False

Question 2 East-west traffic refers to network traffic that travels between servers within the same datacenter.

A True B False

Question 3 Docker and Kubernetes are two technology examples supporting para-virtualization.

A True B False

Question 4 3-tier datacenter networks consist of three distinct layers named core, aggregation, and access.

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Question 8 Replicating data across availability zones helps ensure data durability in case of local failures.

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Question 9 Warehouse-scale computers are single servers that are designed to be used in large-scale data centers.

A True B False

Question 10 Open-loop cooling exploits favourable outdoor conditions to reduce mechanical cooling load.

A True B False

Exercises

Correct answer: +2, No answer: 0.

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A data center hosts a critical service that depends on a system with three components:

- Component A is a central control unit (such as a network controller or orchestrator) essential for system operation. If A fails, the service becomes unavailable.
- Components B1 and B2 are redundant processing units (like application servers) configured so that only one needs to be operational for the system to work correctly.

We know that the reliability of A after 3 years is 0.6, while its MTTR (Mean Time to Repair) is 21 days. Moreover, we know the Availability of B1 and B2, which are respectively 0.65 and 0.8. Compute the availability of component A and compute the overall system availability.

Question 13
Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 420 days and a Mean Time To Repair (MTTR) of 7 days. What is the maximum number of disks that can be included while ensuring that the overall storage ($MTTF_{RAID1+0}$) is at least 6 years

Question 14
An Internet based system includes initially a Web server, an application server, and a DBMS. After monitoring the system for $T = 1800s$ you get the following data:

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Question 15
Given the system described at Question 14, you expect the incoming load to grow to $\lambda = 5req/s$. Determine the minimum number of servers at each physical layer in a way that the utilization of each server is at most 55% (assume that the servers of the same type evenly distribute their load).

Question 16
Assume you upgraded the system according to your answer to Question 15. What is the maximum throughput that the new system can achieve?

Open Questions

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Question 17

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

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Answer Sheets (Page 1)

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Computing Infrastructures - July 4, 2025

Answer Sheets (Page 2)

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Answer Sheets (Page 3)

Student ID (Codice Persona): ....

True/False Questions

Question 01 : [x] A [ ] B

Question 02 : [x] A [ ] B

Question 03 : [ ] A [x] B

Question 04 : [x] A [ ] B

Question 05 : [x] A [ ] B

Question 06 : [x] A [ ] B

Question 07 : [ ] A [x] B

Question 08 : [x] A [ ] B

Question 09 : [ ] A [x] B

Question 10 : [x] A [ ] B

Exercises

Question 11 : 32 SERVERS

Question 12 : A.A = 0,9803 Av.Sys = 0,9210

Question 13 : 11,5 ⇒ 10 DISKS

Question 14 : D_{WEB} = 2,25 sec D_{App} = 1,5 sec D_{DBMS} = 1,8 sec

Question 15 : N_{WEB} = 21 N_{App} = 16 N_{DBMS} = 17

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Question 1 Cloud-native applications are designed to run in the cloud and often use microservices and containers.

A False   B True

Question 2 Replicating data across availability zones helps ensure data durability in case of local failures.

A True   B False

Question 3 Open-loop cooling exploits favourable outdoor conditions to reduce mechanical cooling load.

A True   B False

Question 4 Warehouse-scale computers are single servers that are designed to be used in large-scale data centers.

A True   B False

Question 5 All hardware accelerators used in the datacenter context use seamless programming interfaces and software frameworks, and can be easily interchanged.

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Question 6 3-tier datacenter networks consist of three distinct layers named core, aggregation, and access.

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Question 7 Docker and Kubernetes are two technology examples supporting para-virtualization.

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Question 8 The average seek time of an HDD can be estimated as one-third of the time it takes the actuator arm to move from the innermost to the outermost track.

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Question 9 East-west traffic refers to network traffic that travels between servers within the same datacenter.

A True   B False

Question 10 Data centers using HDDs typically ignore redundancy features, assuming disk failures are very rare.

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- DBMS utilization $U_{DBMS} = 60\%$

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Answer Sheets (Page 2)

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Answer Sheets (Page 3)

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True/False Questions

Question 01 : ☐ A ☒ B

Question 02 : ☒ A ☐ B

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Question 04 : ☐ A ☒ B

Question 05 : ☐ A ☒ B

Question 06 : ☒ A ☐ B

Question 07 : ☒ A ☐ B

Question 08 : ☐ A ☒ B

Question 09 : ☒ A ☐ B

Question 10 : ☐ A ☒ B

Exercises

Question 11 : 32 SERVERS

Question 12 : $A_A = 0,9957$ $A_{SYS} = 0,9584$

Question 13 : 15,1 => 14 DISKS

Question 14 : $D_{WEB} = 4,5$ SEC $D_{APP} = 3$ SEC $D_{DBMS} = 3,6$ SEC

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- Components B1 and B2 are redundant processing units (like application servers) configured so that only one needs to be operational for the system to work correctly.

We know that the reliability of A after 3 years is 0.6, while its MTTR (Mean Time to Repair) is 21 days. Moreover, we know the Availability of B1 and B2, which are respectively 0.65 and 0.8. Compute the availability of component A and compute the overall system availability.

Question 13
Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 420 days and a Mean Time To Repair (MTTR) of 4 days. What is the maximum number of disks that can be included while ensuring that the overall storage ($MTTF_{RAID1+0}$) is at least 8 years

Question 14
An Internet based system includes initially a Web server, an application server, and a DBMS. After monitoring the system for $T = 1800s$ you get the following data:

- Number of completions $C = 300$
- Web server utilization $U_{Web} = 75\%$
- Application server utilization $U_{App} = 50\%$
- DBMS utilization $U_{DBMS} = 60\%$

What are the service demand at the three servers? What is the service demand for the three servers (i.e. D_{Web}, D_{App}, D_{DBMS})?

Question 15
Given the system described at Question 14, you expect the incoming load to grow to $\lambda = 3req/s$. Determine the minimum number of servers at each physical layer in a way that the utilization of each server is at most 70% (assume that the servers of the same type evenly distribute their load).

Question 16
Assume you upgraded the system according to your answer to Question 15. What is the maximum throughput that the new system can achieve?

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.

Computing Infrastructures - July 4, 2025

Answer Sheets (Page 1)

First Name (CAPITAL LETTERS): ....

Last Name (CAPITAL LETTERS): ....

Student ID (Codice Persona): ....

Question 17

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Computing Infrastructures - July 4, 2025

Answer Sheets (Page 2)

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

Computing Infrastructures - July 4, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona): ....

True/False Questions

Question 01 : [x] A [ ] B

Question 02 : [ ] A [x] B

Question 03 : [ ] A [x] B

Question 04 : [ ] A [x] B

Question 05 : [x] A [ ] B

Question 06 : [x] A [ ] B

Question 07 : [x] A [ ] B

Question 08 : [x] A [ ] B

Question 09 : [x] A [ ] B

Question 10 : [x] A [ ] B

Exercises

Question 11 : 32 SERVERS

Question 12 : $A_A = 0{,}9903$ $A_{SYS} = 0{,}9210$

Question 13 : 15.1 => 14 DISKS

Question 14 : $D_{WEB} = 4{,}5$ SEC $D_{APP} = 3$ SEC $D_{DBMS} = 3{,}6$ SEC

Question 15 : $N_{WEB} = 20$ $N_{APP} = 13$ $N_{DBMS} = 16$

Question 16 : $X_{MAX} = 4{,}\bar{3}\ \frac{REQ}{SEC}$

Computing Infrastructures
July 4, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona): ....

Last Name: ....
(LAST NAME IN CAPITAL LETTERS)

First Name: ....
(FIRST NAME IN CAPITAL LETTERS)

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If needed, you can use this page for notes. Any answer written here will be ignored.

Computing Infrastructures
July 4, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona): ....

Last Name: ....
(LAST NAME IN CAPITAL LETTERS)

First Name: ....
(FIRST NAME IN CAPITAL LETTERS)

Exam Duration: 1hour and 30min

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Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

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Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the *Open Questions* should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points**. An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.

True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 The average seek time of an HDD can be estimated as one-third of the time it takes the actuator arm to move from the innermost to the outermost track.

A True B False

Question 2 Open-loop cooling exploits favourable outdoor conditions to reduce mechanical cooling load.

A True B False

Question 3 All hardware accelerators used in the datacenter context use seamless programming interfaces and software frameworks, and can be easily interchanged.

A False B True

Question 4 3-tier datacenter networks consist of three distinct layers named core, aggregation, and access.

A True B False

Question 5 Replicating data across availability zones helps ensure data durability in case of local failures.

A True B False

Question 6 East-west traffic refers to network traffic that travels between servers within the same datacenter.

A False B True

Question 7 Cloud-native applications are designed to run in the cloud and often use microservices and containers.

A True B False

Question 8 Data centers using HDDs typically ignore redundancy features, assuming disk failures are very rare.

A False B True

Question 9 Warehouse-scale computers are single servers that are designed to be used in large-scale data centers.

A False B True

Question 10 Docker and Kubernetes are two technology examples supporting para-virtualization.

A True B False

Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 8 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?

Question 12

A datacenter hosts a critical service that relies on a system composed of three components:

- Component A is a central control unit (e.g., a network controller or orchestrator) that is mandatory for system operation. If A fails, the service becomes unavailable.
- Components B1 and B2 are redundant processing units (e.g., heterogeneous application servers) configured such that only one needs to be operational for the system to function correctly.

We know that the reliability of A after 2 years is 0.8, while its MTTR (Mean Time to Repair) is 14 days. Moreover, we know the Availability of B1 and B2, which are respectively 0.75 and 0.85. Compute the availability of component A and compute the overall system availability.

Question 13
Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 420 days and a Mean Time To Repair (MTTR) of 4 days. What is the maximum number of disks that can be included while ensuring that the overall storage ($MTTF_{RAID1+0}$) is at least 8 years

Question 14
An Internet-based system includes initially a Web server, an application server, and a DBMS. After monitoring the system for $T = 1800s$, you get the following data:

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- Web server utilization $U_{Web} = 75\%$
- Application server utilization $U_{App} = 50\%$
- DBMS utilization $U_{DBMS} = 60\%$

What is the service demand for the three servers (i.e. D_{Web}, D_{App}, D_{DBMS})?

Question 15
Given the system described at Question 14, you expect the incoming load to grow to $\lambda = 3req/s$. Determine the minimum number of servers at each physical layer in a way that the utilization of each server is at most 55% (assume that the servers of the same type evenly distribute their load).

Question 16
Assume you upgraded the system according to your answer to Question 15. What is the maximum throughput that the new system can achieve?

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

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Computing Infrastructures - July 4, 2025

Answer Sheets (Page 1)

First Name (CAPITAL LETTERS): ....

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Question 17

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

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Computing Infrastructures - July 4, 2025

Answer Sheets (Page 2)

Question 18

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Computing Infrastructures - July 4, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona): ....

True/False Questions

	A	B
Question 01 :	☒	☐
Question 02 :	☒	☐
Question 03 :	☒	☐
Question 04 :	☒	☐
Question 05 :	☒	☐
Question 06 :	☐	☒
Question 07 :	☒	☐
Question 08 :	☒	☐
Question 09 :	☒	☐
Question 10 :	☐	☒

Exercises

Question 11 : 32 SERVERS

Question 12 : $A_{A} = 0{,}9957$ $A_{SYS} = 0{,}9584$

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Question 16 : $X_{MAX} = 5{,}5$ REQ/SEC

Computing Infrastructures
July 4, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

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Computing Infrastructures
July 4, 2025

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A False　　B True

Question 2 Replicating data across availability zones helps ensure data durability in case of local failures.

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Question 13
Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 520 days and a Mean Time To Repair (MTTR) of 7 days. What is the maximum number of disks that can be included while ensuring that the MTTF for overall storage ($MTTF_{RAID1+0}$) is at least 8 years

Question 14
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Computing Infrastructures - July 4, 2025

Answer Sheets (Page 1)

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Computing Infrastructures - July 4, 2025

Answer Sheets (Page 2)

Question 18

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+6/9/2+

Computing Infrastructures - July 4, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona): ....

True/False Questions

Question 01 : ☐A ☒B

Question 02 : ☒A ☐B

Question 03 : ☐A ☒B

Question 04 : ☒A ☐B

Question 05 : ☐A ☒B

Question 06 : ☐A ☒B

Question 07 : ☐A ☒B

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Question 10 : ☐A ☒B

Exercises

Question 11 : 32 SERVERS

Question 12 : $A_{WA} = 0{,}9957$ $A_{SYS} = 0{,}9584$

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Question 14 : $D_{WEB} = 4.5$ sec $D_{APP} = 3$ sec $D_{DBMS} = 3.6$ sec

Question 15 : $N_{WEB} = 25$ $N_{APP} = 17$ $N_{DBMS} = 20$

Question 16 : $X_{MAX} = 5{,}\bar{5}\ \frac{REQ}{SEC}$

Computing Infrastructures
July 4, 2025

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Student ID (Codice Persona): ....

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